Lecture 7: Techniques of Integration

IV. The method of Partial Fractions, part I

ex.1. $\displaystyle\int \frac{2x+1}{x^2+x-2}\,dx = \int \frac{1}{u}\,du = \ln|x^2+x-2| + c$

(u-sub: $u = x^2+x-2$ and $du = (2x+1)dx$)

ex.2. $\displaystyle\int \frac{x+5}{x^2+x-2}\,dx = ?$ Partial fraction decomposition.

$$\left(\frac{x+5}{x^2+x-2} = \frac{2(x+2)-(x-1)}{(x-1)(x+2)} = \frac{2}{x-1} - \frac{1}{x+2}\right)$$

← simplify / combine

$$\int \frac{x+5}{x^2+x-2}\,dx = \int \frac{2}{x-1} - \frac{1}{x+2}\,dx$$

$$= 2\ln|x-1| - \ln|x+2| + c$$

The method of Partial Fractions allows us to integrate a rational function, $P(x)/Q(x)$, by first expressing it as a sum of simpler fractions called **Partial Fractions** that can be integrated easily.

The process of breaking down a proper rational function $p(x)/q(x)$ into smaller partial fractions is called the **Partial Fraction Decomposition.**

Proper rational expression:

Let $f(x) = \dfrac{P(x)}{Q(x)}$ where both $P(x)$ and $Q(x)$ are polynomials and the degree of $P(x)$ is strictly smaller than the degree of $Q(x)$, then this rational expression is proper. $\deg(P) < \deg(Q)$

Partial fractions can only be done if the rational expression is proper. **

Partial Fraction Decomposition

To decompose a rational expression **is to break up** a proper fraction into sums of smaller partial fractions, (ie. undo the above):

1. factor the denominator as far as possible.
2. every factor of the denominator gets a term.

Four possible cases of partial fraction decomposition:

Fundamental Thm of Algebra, states that every Polynomial $Q(x)$with real coefficients can be written as a product of linear $(ax+b)$ and irreducible quadratic factors (ax^2+bx+c) with real coefficients. $b^2-4ac<0$

$x^3-1=(x-1)(x^2+x+1)$

irreducible

$x^2+2x+1=(x+1)^2$ reducible

$x^2-x-2=(x-2)(x+1)$

$(b^2-4ac>0)$

	Factors in denominator	Forms in partial fraction decomposition
L7	$ax+b$	$\dfrac{A}{ax+b}$
	$(ax+b)^n$	$\dfrac{A_1}{ax+b}+\cdots+\dfrac{A_n}{(ax+b)^n},$
L8 and L9	ax^2+bx+c	$\dfrac{Ax+B}{ax^2+bx+c}$
	$(ax^2+bx+c)^n$	$\dfrac{A_1x+B_1}{ax^2+bx+c}+\cdots+\dfrac{A_nx+B_n}{(ax^2+bx+c)^n},$

We start with the simplest case:

Case 1 $Q(x)$ is a product of **distinct linear factors with no repeats.**

$$Q(x) = (a_1x + b_1)\cdots(a_nx + b_n)$$

then, there is a **partial fraction decomposition (PFD)**

$$f(x) = \frac{P(x)}{Q(x)} = \frac{A_1}{a_1x + b_1} + \cdots + \frac{A_n}{a_nx + b_n}$$

for suitable constants $A_1, \cdots A_n$.

Once we find the PFD, we can integrate the rational function $f(x)$ easily.

ex. 1. First, practice writing out the **form** of the partial fraction **decomposition**.
(Do not determine the constants yet.)

proper ✓

1. $\dfrac{3x}{2x^2 + x - 3} = \dfrac{3x}{(2x+3)(x-1)} = \dfrac{A}{2x+3} + \dfrac{B}{x-1}$

proper ✗ ↓ long division

2. $\dfrac{x^2 + 1}{x^2 - 1} = 1 + \dfrac{2}{x^2-1}$ $\left(1 + \frac{A}{x-1} + \frac{B}{x+1}\right)$ $(x+1)(x-1)$

$$\begin{array}{r} 1 \\ x^2+0-1 \overline{\big) x^2+0+1} \\ -)\ x^2+0-1 \\ \hline 2 \end{array}$$

$= 1 + \dfrac{A}{x+1} + \dfrac{B}{x-1}$

Evaluating the following integrals:

ex. 2. $\displaystyle\int \frac{1}{x^2+3x-4}\,dx.$ (factor denominator)

PFD:

$$\left(\frac{1}{(x+4)(x-1)} = \frac{A}{x+4} + \frac{B}{x-1}\right)(x-1)\cancel{(x+4)} \quad (\star)$$

Finding the Constants:

1. Multiply both sides by $(x+4)$ to clear $A's$ denominators:

$$\frac{1}{x-1} = A + \cancel{\frac{B}{x-1}(x+4)} \to 0$$

Let $\boxed{x=-4,}$ $\quad \dfrac{1}{-5} = A \Longrightarrow A = -\dfrac{1}{5}$

2. Multiply both sides by $(x-1)$ to clear $B's$ denominators:

$$\frac{1}{x+4} = \cancel{\frac{A}{x+4}(x-1)} \to 0 + B$$

Let $\boxed{x=1,}$ $\quad \dfrac{1}{5} = B$

$\boxed{\textbf{Evaluate}}$:

$$\int \frac{1}{(x+4)(x-1)}dx = \frac{1}{5}\int \left(\frac{1}{x-1} - \frac{1}{x+4}\right)\ dx.$$

$$= \frac{1}{5}(\ln|x-1| - \ln|x+4|) + C$$

$$\text{or } = \frac{1}{5}\ln\left|\frac{x-1}{x+4}\right| + c$$

There is a quicker, more clever way to determine these constants in ($\star$) by using the 'cover-up' method.

$\boxed{\textbf{Cover-Up Method}}$ to evaluate the constants:

<u>To find A:</u>

Cover everything but A on the right, and $(x+4)$ on the left, plug in $x = -4$ to both sides, $\rightarrow A = -\frac{1}{5}$.

<u>To find B:</u>

Cover everything but B on the right, and $(x-1)$ on the left, plug in $x = 1$ to both sides, $\rightarrow B = \frac{1}{5}$.

$a \neq 0$ $\quad \int \frac{1}{ax-b}\,dx = \frac{1}{a}\ln|ax-b| + C$

$(u = ax-b)$

Recap the steps for <u>case 1</u>:

1. Ensure $f(x)$ is **Proper**(use long division if not)
2. **Factor** $Q(x)$ as far as possible
3. Do **Partial Fraction Decomposition** (PFD)
4. Find the **Contants** using **'Cover Up'**
5. Integrate the Partial Fractions.

<u>ex.</u> 3. $\displaystyle\int \frac{4y^2 - 7y - 12}{y(y+2)(y-3)}\,dy$

(check: proper and $Q(x)$ factored)

$\boxed{\textbf{PFD}}$: $\displaystyle\frac{4y^2 - 7y - 12}{y(y+2)(y-3)} = \frac{A}{y} + \frac{B}{y+2} + \frac{C}{y-3}$

$\boxed{\textbf{Constants}}$: (use **cover-up** method:)

Let $y = 0$, $\quad A = \dfrac{0 - 0 - 12}{2\cdot(-3)} = \dfrac{-12}{-6} = 2$

Let $y = -2$, $\quad B = \dfrac{4(-2)^2 - 7(-2) - 12}{(-2)\cdot(-5)} = \dfrac{9}{5}$

Let $y = 3$, $\quad C = \dfrac{4\cdot 3^2 - 7\cdot 3 - 12}{3\cdot 5} = \dfrac{1}{5}$

$\boxed{\textbf{Evaluate}}$:

$$\int \frac{4y^2 - 7y - 12}{y(y+2)(y-3)}dy = \int \left(\frac{2}{y} + \frac{9/5}{y+2} + \frac{1/5}{y-3}\right) dy$$

$$= 2\ln|y| + \frac{9}{5}\ln|y+2| + \frac{1}{5}\ln|y-3| + C$$

NYTI: Evaluate the integral $\displaystyle\int_1^2 \frac{4y^2 - 7y - 12}{y(y+2)(y-3)}dy$

$(= \frac{9}{5}(\ln(8/3)))$.

$$= 2\ln|y| + \frac{9}{5}\ln|y+2| + \frac{1}{5}\ln|y-3| \Big|_1^2$$

$$= 2\ln 2 + \frac{9}{5}\ln 4 + \cancel{\frac{1}{5}\ln 1}^{\,0} - \left(\cancel{2\ln 1}^{\,0} + \frac{9}{5}\ln 3 + \frac{1}{5}\ln 2\right)$$

$$= 2\ln 2 + \frac{18}{5}\ln 2 - \frac{1}{5}\ln 2 - \frac{9}{5}\ln 3$$

$$= \frac{27}{5}\ln 2 - \frac{9}{5}\ln 3$$

$$= \frac{9}{5}\ln 8 - \frac{9}{5}\ln 3$$

$$= \frac{9}{5}\ln\left(\frac{8}{3}\right)$$

Case 2 $Q(x)$ has **repeated linear factors**

$$(ax+b)^n,$$

then, there is a PFD form:

$$\frac{P(x)}{Q(x)} = \frac{A_1}{ax+b} + \frac{A_2}{(ax+b)^2} + \cdots + \frac{A_n}{(ax+b)^n}$$

for suitable constants $A_1, \cdots A_n$.

Practice writing out the **form** of the partial fraction decomposition for case 2.
(Do not determine the constants yet.)

1. $\dfrac{x^2+9x-12}{(3x-1)(x+6)^2} = \dfrac{A}{3x-1} + \dfrac{B}{x+6} + \dfrac{C}{(x+6)^2}$

NYTI: $\dfrac{1}{x^4-x^3}$ $\left(\frac{A}{x} + \frac{B}{x^2} + \frac{C}{x^3} + \frac{D}{x-1}\right)$

$$= \frac{1}{x^3(x-1)} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x^3} + \frac{D}{x-1}$$

ex. 4. $\displaystyle\int \frac{1}{(x+5)^2(x-1)}dx.$

(check: proper and $Q(x)$ factored)

PFD:

cover-up

$$\frac{1}{(x+5)^2(x-1)} = \frac{A}{x+5} + \frac{B}{(x+5)^2} + \frac{C}{x-1}$$

Constants:

Cover-Up: let $x = 1$, $C = \frac{1}{(1+5)^2} = \frac{1}{36}$

let $x = -5$, $B = \frac{1}{-5-1} = -\frac{1}{6}$

A **can't** be found using 'cover-up'. Instead, plug in a suitable constant, say $x = 0$ to both sides,

$A = -\frac{1}{36}$

Evaluate:

u-sub $u = x+5$

$-\frac{1}{6}\int \frac{1}{u^2}du$

$$\int \frac{1}{(x+5)^2(x-1)}dx = \int \frac{-\frac{1}{36}}{x+5} + \frac{-\frac{1}{6}}{(x+5)^2} + \frac{\frac{1}{36}}{x-1} \quad dx.$$

$$= -\frac{1}{36}\ln|x+5| + \frac{1}{6}\left(\frac{1}{x+5}\right) + \frac{1}{36}\ln|x-1| + C$$

or $= \frac{1}{36}\ln\left|\frac{x-1}{x+5}\right| + \frac{1}{6}\left(\frac{1}{x+5}\right) + c$

NYTI: $\displaystyle\int \frac{ds}{s^2(s-1)^2}$ (ck: proper and Q factored)

$\boxed{\textbf{PFD}}$: $\displaystyle\frac{1}{s^2(s-1)^2} = \frac{A}{s} + \frac{B=1}{s^2} + \frac{C}{s-1} + \frac{D=1}{(s-1)^2}$

$\boxed{\textbf{Contants}}$:(which ones can be found using cover-up?) B and D

Cover-Up: let $s = 0$, $B = \frac{1}{(0-1)^2} = 1$

let $s = 1$, $D = \frac{1}{1^2} = 1$

To find the other coefficients:

$S=-1$: $\frac{1}{(-1)^2(-2)^2} = \frac{A}{-1} + \frac{1}{(-1)^2} + \frac{C}{-2} + \frac{1}{(-2)^2}$

$\left(\frac{1}{4} = -A + 1 - \frac{C}{2} + \frac{1}{4}\right)\cdot 4$

$1 = -4A + 4 - 2C + 1$

$4A + 2C = 4 \longrightarrow 2A + C = 2$ — ①

$S=2$: $\frac{1}{2^2 \cdot 1^2} = \frac{A}{2} + \frac{1}{2^2} + \frac{C}{1} + \frac{1}{1^2}$

$\left(\frac{1}{4} = \frac{A}{2} + \frac{1}{4} + C + 1\right)\cdot 4$

$1 = 2A + 1 + 4C + 4$

$-4 = 2A + 4C \longrightarrow A + 2C = -2$ — ②

Solve equations ① and ② to get $A = 2$ and $C = -2$.

$\boxed{\textbf{Evaluate}}$:

$$\int \frac{ds}{s^2(s-1)^2} = \int \frac{2}{s} + \frac{1}{s^2} + \frac{-2}{s-1} + \frac{1}{(s-1)^2}\, ds.$$

$$= 2\ln|s| - \frac{1}{s} - 2\ln|s-1| - \frac{1}{s-1} + C$$

or $= 2\ln\left|\frac{s}{s-1}\right| - \frac{1}{s} - \frac{1}{s-1} + c$